

“Initial Plan: Exploring Large Language Models for Enhanced Game Experiences”

Authors: Isaac Eccleston, Frank Langbein

Project Description

Life-simulation games, such as *Stardew Valley*, are games that aim to provide engaging and immersive worlds for their players. One of the key factors to this experience is the player’s interactions with in-game characters. These non-player characters (NPCs) provide context, movement, and a sense of life to the world that they inhabit, meaning they greatly deepen the player’s emotional investment during narrative-focused gameplay. Despite their importance, many NPCs in contemporary life-simulation games have a shallow depth of interactions with the player, their dialogue and events tightly scripted and the nature of their relationship with the player overly discrete. This results in limited emotional investment, repetitive NPC experiences on repeat plays, and a reduced sense of immersion in the world we want to create.

The problem that my project aims to address is this lack of depth and adaptability in player-NPC interactions and relationships in life-simulation games. The common approach of dialogue trees becomes infeasible as the intricacy of the world develops, and therefore relationships with the player are often reduced to simple states (e.g. ‘pleased’ vs ‘displeased’). This can lead to NPCs not bearing realistic long-term emotional impact from player-caused conflicts in their world, which reduces their agency and their believability as real characters in a narrative setting. These long-term changes and developments are key to human social interaction, and their absence from many NPC systems weakens the strength of both their narrative and gameplay utility.

My proposed solution is to explore the use of Large Language Models (LLMs) to enhance NPC interactions and their effect on the player’s experience, within a controlled game environment. The focus is the development of a small scope life-simulation prototype that functions as the basis for an NPC dialogue and player interaction system. Within this created world, NPCs will have their emotions, motivations, and activities tracked and represented in a structured state, as well as knowledge of the player’s actions and progression. This information will then be used to engineer responses through the use of LLMs upon player-NPC interaction, giving the prototype dynamic character dialogue that relies on both world and player context.

Whilst aiming to generate all NPC-player interactions through LLMs, the system will have the necessary fallback dialogue trees to continue narratives in the case of incoherent or inappropriate responses, ensuring gameplay fidelity and consistency. Responses will also be checked against the NPCs current model to ensure tonal parity and correct context. By introducing conflict events to explore player-NPC(s) interactions, the project will explore how dynamically generated dialogue can reflect greater emotional depth than traditional dialogue trees.

By limiting the system to a prototype of NPC-interactions in a small setting, the project remains focused on evaluating LLM generated responses and avoids unnecessary development on narrative or aesthetic. LLM responses are restricted to player interactions, which reduces the frequency of usage and associated costs / latency, whilst also ensuring that their output is manageable, able to be sanitised, and gameplay relevant. Having an underlying NPC modelling system allows for evaluation of LLM generated dialogue compared to traditional dialogue trees and demonstrates the project's suitability. Overall, my solution provides a practical approach to exploring the enhancement on emotional investment and player opinion of NPC dialogue through the use of LLMs.

Aims and Objectives

My overall aim is:

- To explore how Large Language Models can be used to simulate believable NPC emotional responses and relationship progression within a game engine.

My specific objectives are:

1. Develop a top-down style life-simulation prototype that supports player-NPC interaction and functions as a controlled environment for simulating conflict events and their effects on LLM driven player dialogue and relationships. A potential risk could be excessive focus on non-NPC related gameplay and aesthetics rather than LLM integration, leading to an unfinished dialogue evaluation. To mitigate this, I will restrict the prototype to a limited number of scenes and use open-license art assets to enable the development focus to instead be on the dialogue system and LLM integration.
2. Design and implement a structured NPC model that encompasses emotional state, motivations, and relation to game events, to inform LLMs for generation of coherent and engaging contextual responses. A risk for this objective could be emotional modelling causing inconsistent NPC behaviour, meaning that responses from NPCs may be incoherent and lead to integration difficulties. This will be mitigated by iteratively developing the NPC model structures, ensuring that the relevant information to the player is always delivered and systems are in place to sanitise LLM output before it's shown to the player.
3. Integrate a Large Language Model to generate NPC dialogue and update emotional variables based on gameplay state, player history, and mood. Potential risks for generating LLM responses are high latency and high cost, which may affect gameplay flow or exceed constraints for project resources. I will mitigate this by limiting the usage of LLM calls to player interactions, making sure to construct my prompts in a consistent format that reduces incoherent answers and duplicate calls, and by implementing a skeleton script for fallback dialogue if LLM responses fail.
4. Evaluate the feasibility and effectiveness of an NPC dialogue and relationship system driven by LLM integrations when compared to a standard tree based system. A risk of this objective is that results may be subjective or hard to quantify, I will mitigate this by ensuring I gather quantitative metrics that are relative to my qualitative analysis to provide a strong and useful evaluation.

Feasibility

A key factor affecting the feasibility of my project is the usage of LLMs for generating NPC dialogue. This generation will be done either via subscription to a hosted LLM API, or, if possible, through a locally hosted model. Associated costs with LLM APIs introduce feasibility concerns, and therefore management of LLM calls and reduction of their frequency will be crucial. A locally hosted model would allow for more LLM calls and increased LLM impact on gameplay, but the installation and training of a specific model rather than using an off-the-shelf alternative would be another milestone to consider for the project and may raise issues of deployment and specialised hardware requirements.

In addition, there are ethical feasibility concerns related to players conversing with LLM-driven NPCs, particularly the possibility of inappropriate or unethical content being generated. This risk will be addressed by sanitising all LLM responses prior to player view, using filtering rules and contextual constraints to ensure relevance, appropriateness, and consistency. The project doesn't involve real-world user information, and the prototype takes place in an entirely fictional environment which means that output should not cross any data or ethical boundaries. Ethical approval is therefore not expected to be a serious roadblock, but the systems will be in place to ensure appropriateness.

Furthermore, in aim to reduce scope, I aim to use open-license assets for the artwork for the prototype. If this is possible, all assets will be used within their license agreements and with proper credit. If no assets are available, my own spritework will suffice but will not be a focus of the development.

Work Plan

Week	Tasks	Milestone	Deliverable
1 26/1/26	Write initial plan		
2 2/2/26	- Submit initial plan - Collate relevant literature		Initial Plan - 2/2/26
3 9/2/26	- Review literature relating to traditional tree-based NPC interactions - Review literature relating to generating and reviewing LLM responses - Draft literature review and background section		
4 16/2/26	- Iterate on literature review and background, collating more research - Begin building prototype top-down simulator in game engine		

	Learn relevant LLM integration technologies to develop prototype		
5 23/2/26	- Continue building prototype, narrative systems, leaving space for LLM integration - Write about software development for report draft		
6 2/3/26	- Begin LLM integration & continue prototype development - Continue writing about software development - Iterate on written sections from feedback		
7 9/3/26	- Finish LLM integration and write basic alternate conversation trees - Finish MVP - Find test players and write evaluation rubric / surveys	MVP	
8 16/3/26	- Document player test feedback - Improve prototype based on player feedback - Draft writing about improvements made and evaluations		
9 13/4/26	- Further evaluation of LLM and its effectiveness - Continue writing sections for final report - Final touches to solution before submission	Project solution	
10 20/4/26	Finish draft of final report and get feedback from supervisor	Final report draft	
11 27/4/26	Iterate on final report based on feedback		
12 4/5/26	Submit final report		Final Report – 7/5/26

References

n/a